

Visualising Chance: An Introduction to Probability

Seeing Theory · Brown University

Senior Secondary (Years 11-12)

~30 minutes

Mathematics · Mathematical
Methods / Essential Mathematics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://seeing-theory.brown.edu/basic-probability/index.html>

Curriculum links: probability · random variables · statistical inference · inference

Learning intentions

- I can use simulated repeated trials to estimate the probability of an event.
- I can distinguish between a random variable's theoretical probability and its observed relative frequency.
- I can explain what statistical inference allows us to conclude from sample data.

Getting started

Open Seeing Theory and go to the 'Basic Probability' chapter. Work through the coin flip and card draw demonstrations interactively — run each simulation for a small number of trials first, then increase the number of trials, watching how the results settle toward the theoretical probability.

Tasks

1. Run the coin flip simulation for 10 flips. Record the proportion of heads. Is it exactly 0.5? Should you expect it to be?

2. Now run the same coin flip simulation for 1000 flips. What happens to the proportion of heads compared to your 10-flip result?

3. Explain, in your own words, the difference between the theoretical probability of an event and its observed relative frequency after a set of trials.

4. Run the coin flip simulation for a few different trial sizes and record the proportion of heads observed each time.

Number of trials	Proportion of heads observed

5. Using your table, explain how the Law of Large Numbers is demonstrated by this simulation.

6. If a fair six-sided die is rolled, what is the theoretical probability of rolling a number greater than 4? Use a matching demonstration to run a similar number of trials and compare your simulated result to your calculated answer.

Working:

7. A newspaper reports that 'a new drug worked for 8 out of 10 patients in a trial, so it works for 80% of people.' Using what you've learned about sample size and inference, explain why this conclusion may not be statistically sound.

8. Explain how the ideas in this simulation (repeated trials, relative frequency, sample size) connect to how statisticians estimate probabilities and make inferences in real research.

Extension (optional)

Design your own repeated-trial scenario (e.g. drawing coloured balls from a bag) and use Seeing Theory's tools to estimate its probability from simulated trials. Compare your simulated estimate to the true theoretical probability, and explain any difference you find.
